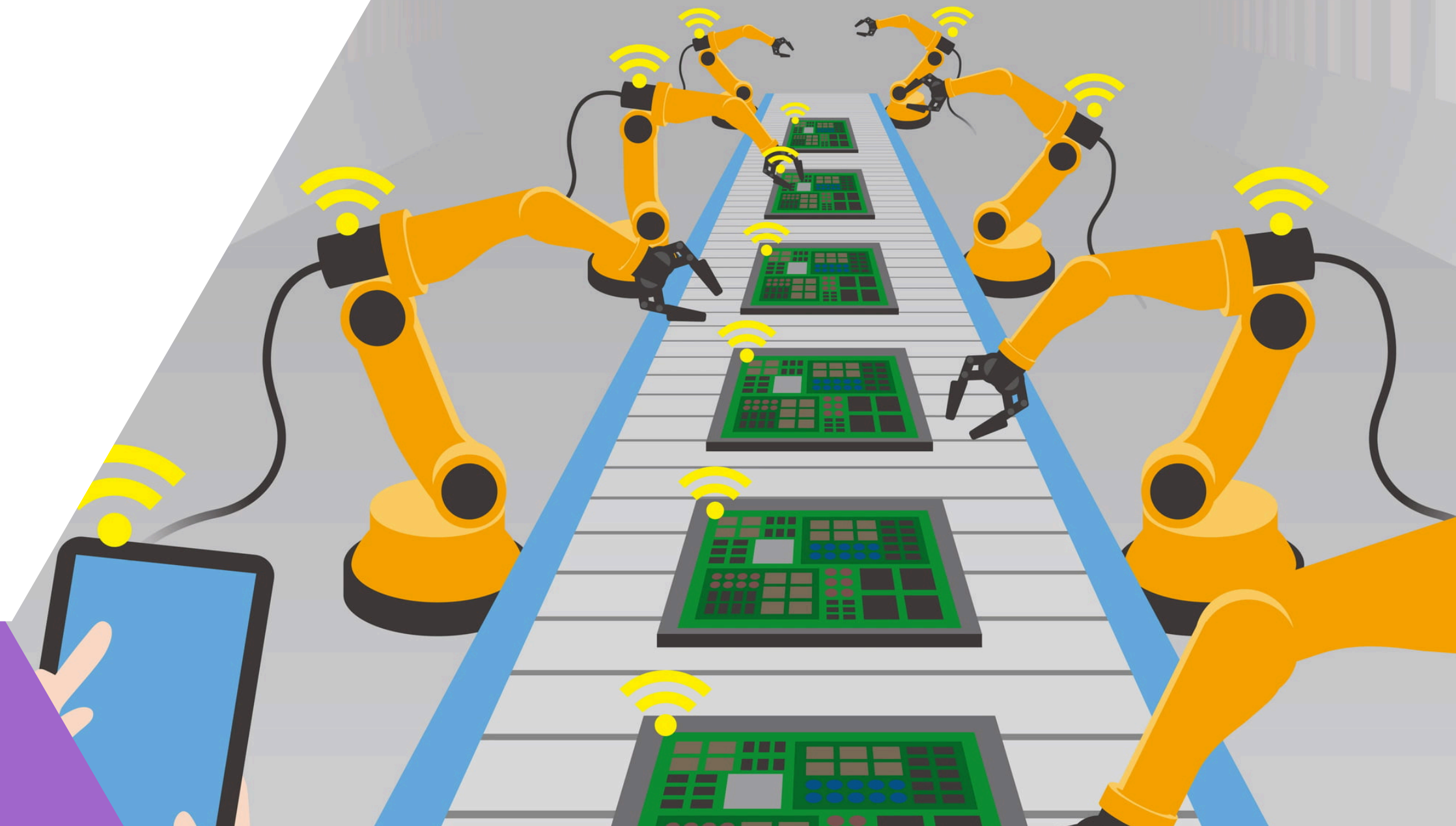




AUTOMATIC ASSEMBLY SYSTEM



Aim & Objective

Aim:

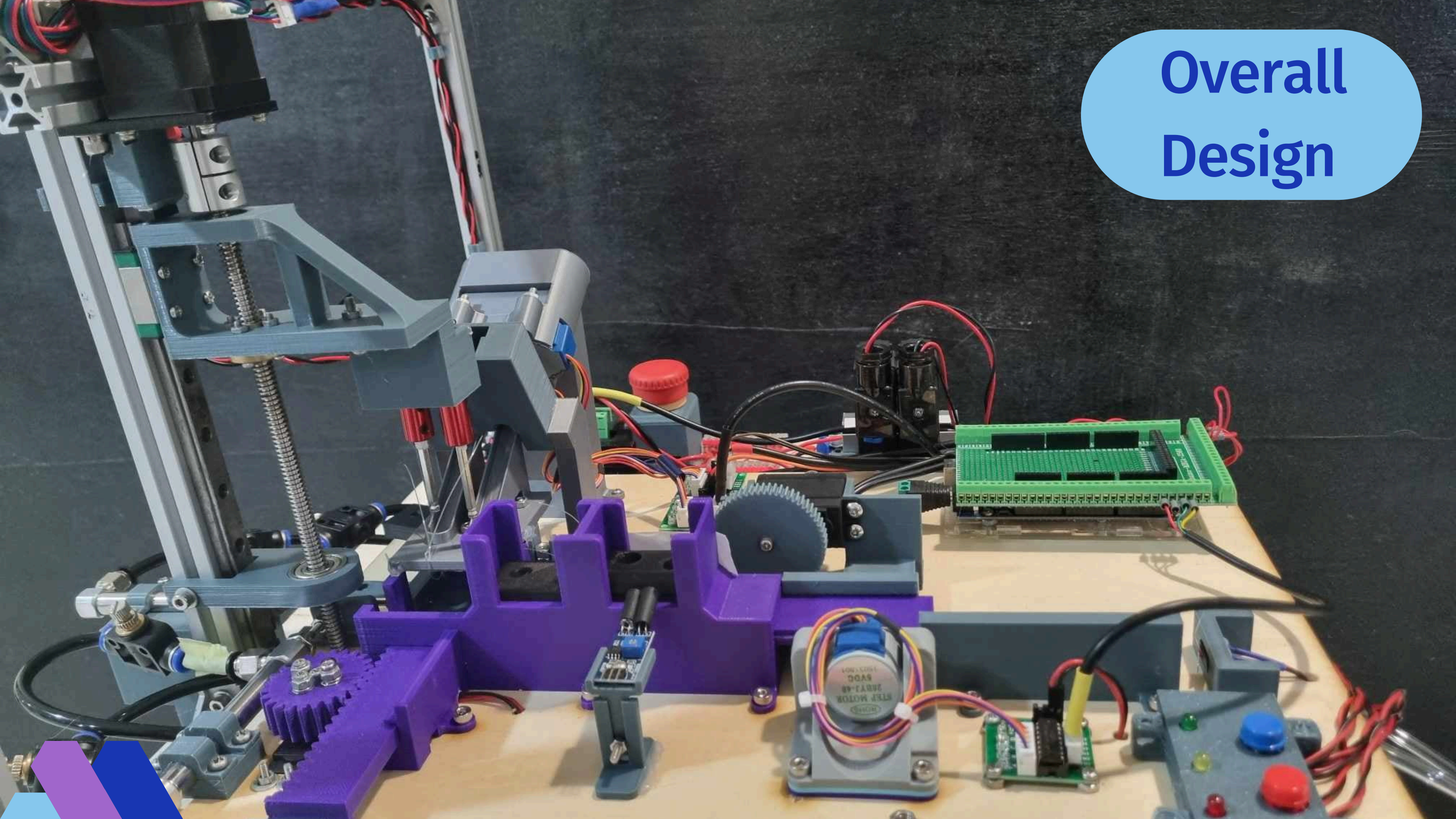
To develop an automated mechatronics assembly system that could assemble 2 blocks together using fasteners within 1 minute.

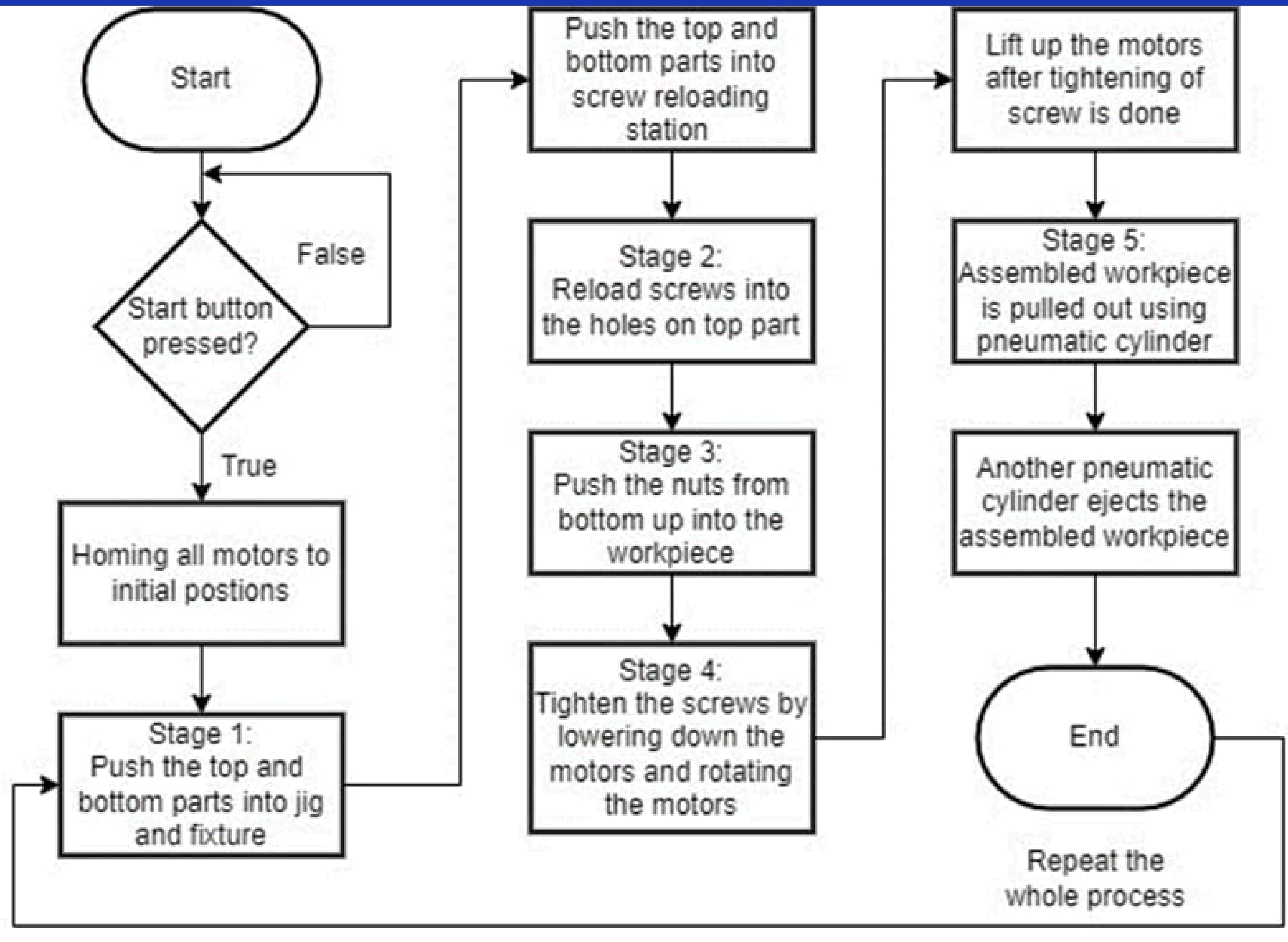
Objective:

- Development of a jig and fixture
- Development of a semi-automated feeding system
- Development of a screwing mechanism



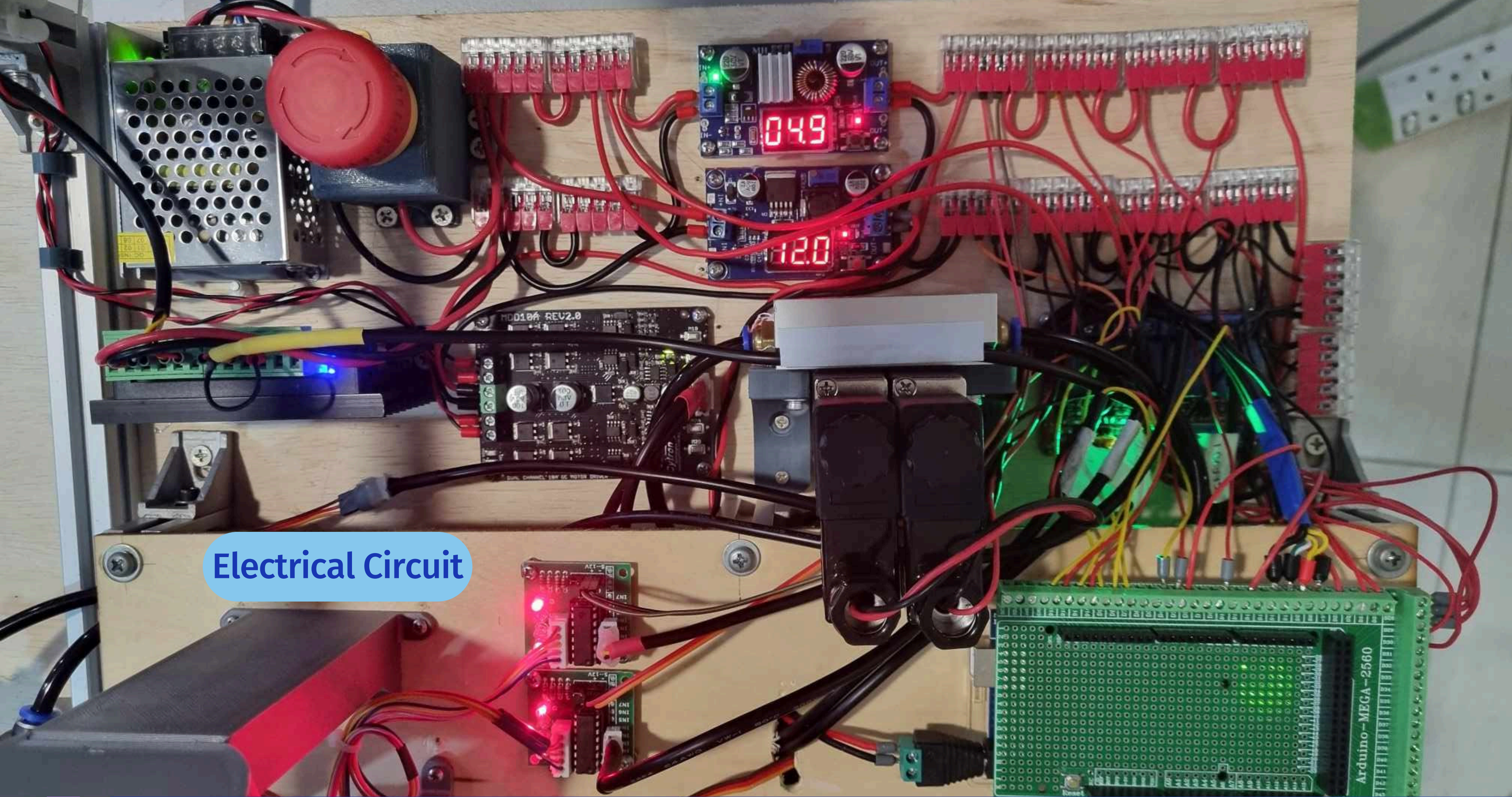
Overall Design





Flowchart of Overall Process

Electrical Circuit



Control Buttons

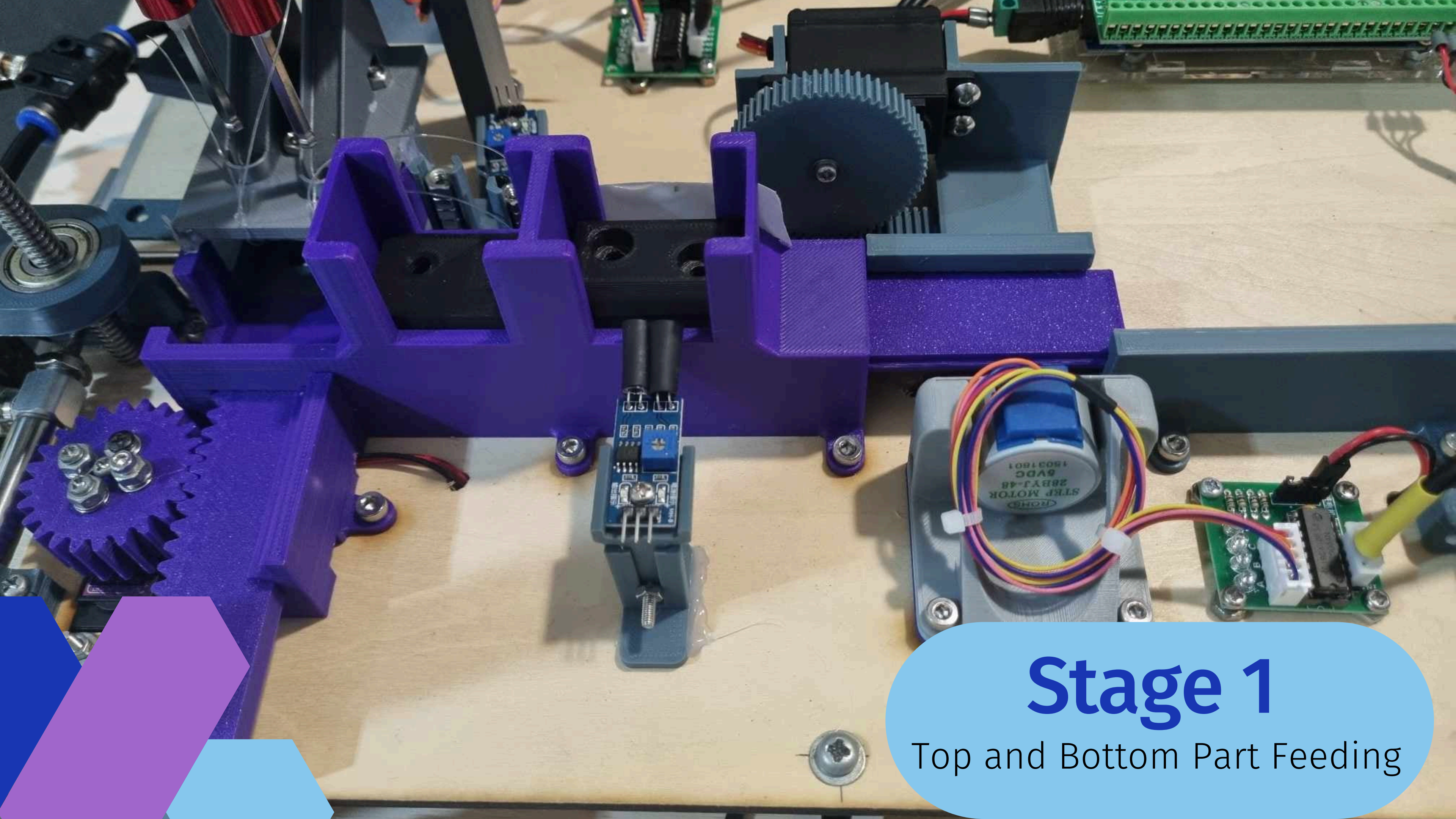
Stop Light

Warning Light

Start Light

Stop Button

Start Button

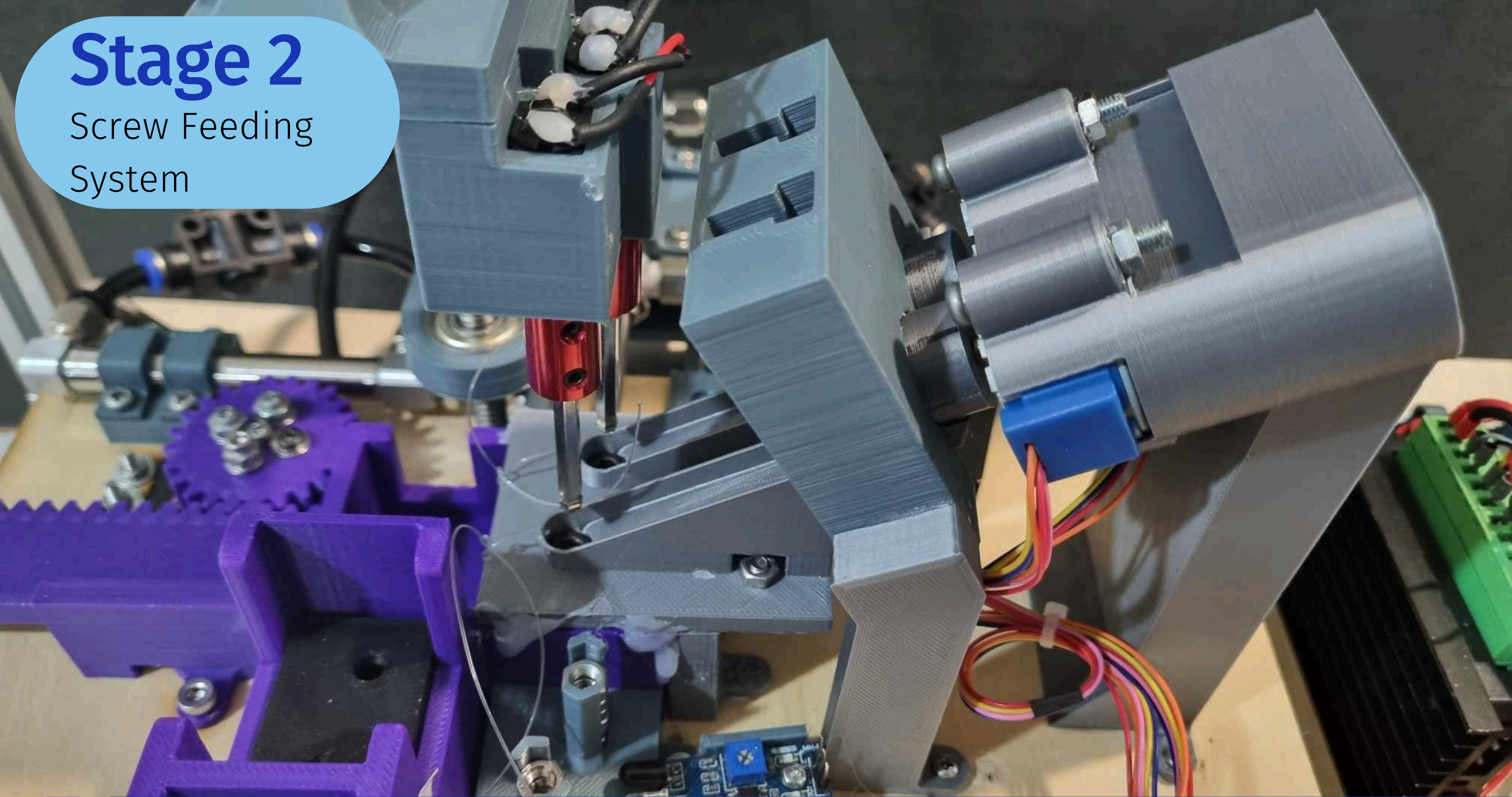


Stage 1

Top and Bottom Part Feeding

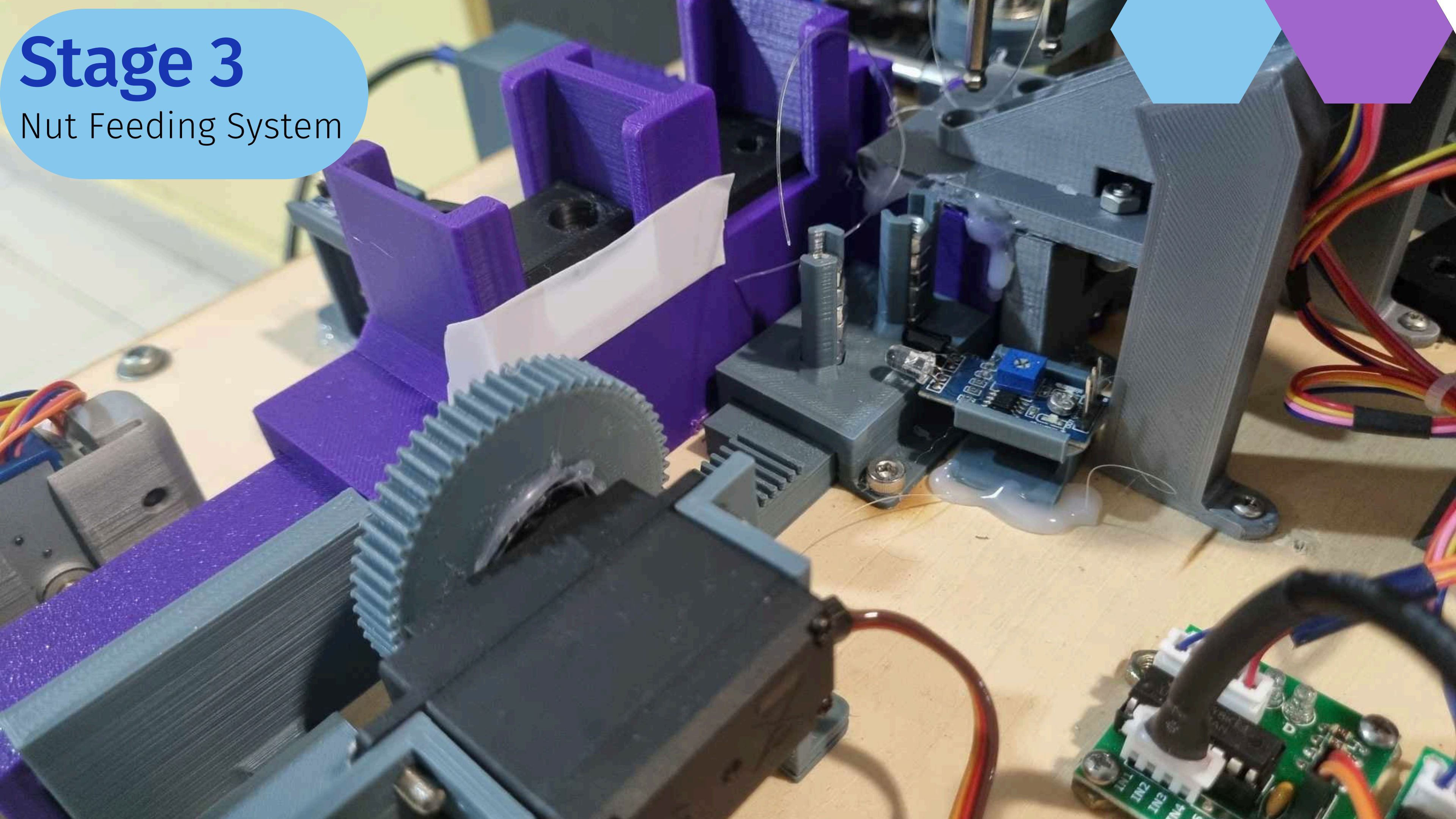
Stage 2

Screw Feeding
System



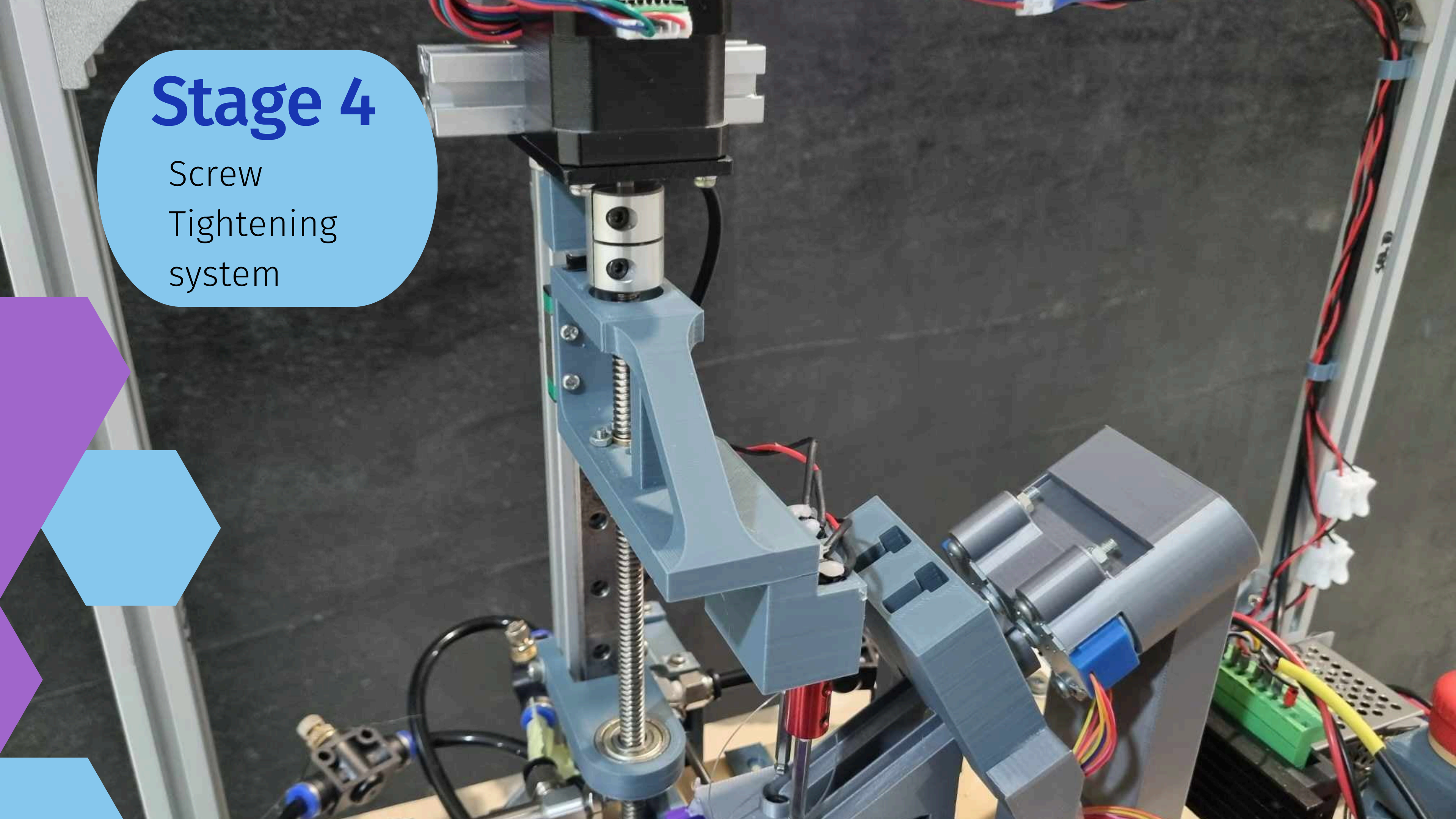
Stage 3

Nut Feeding System



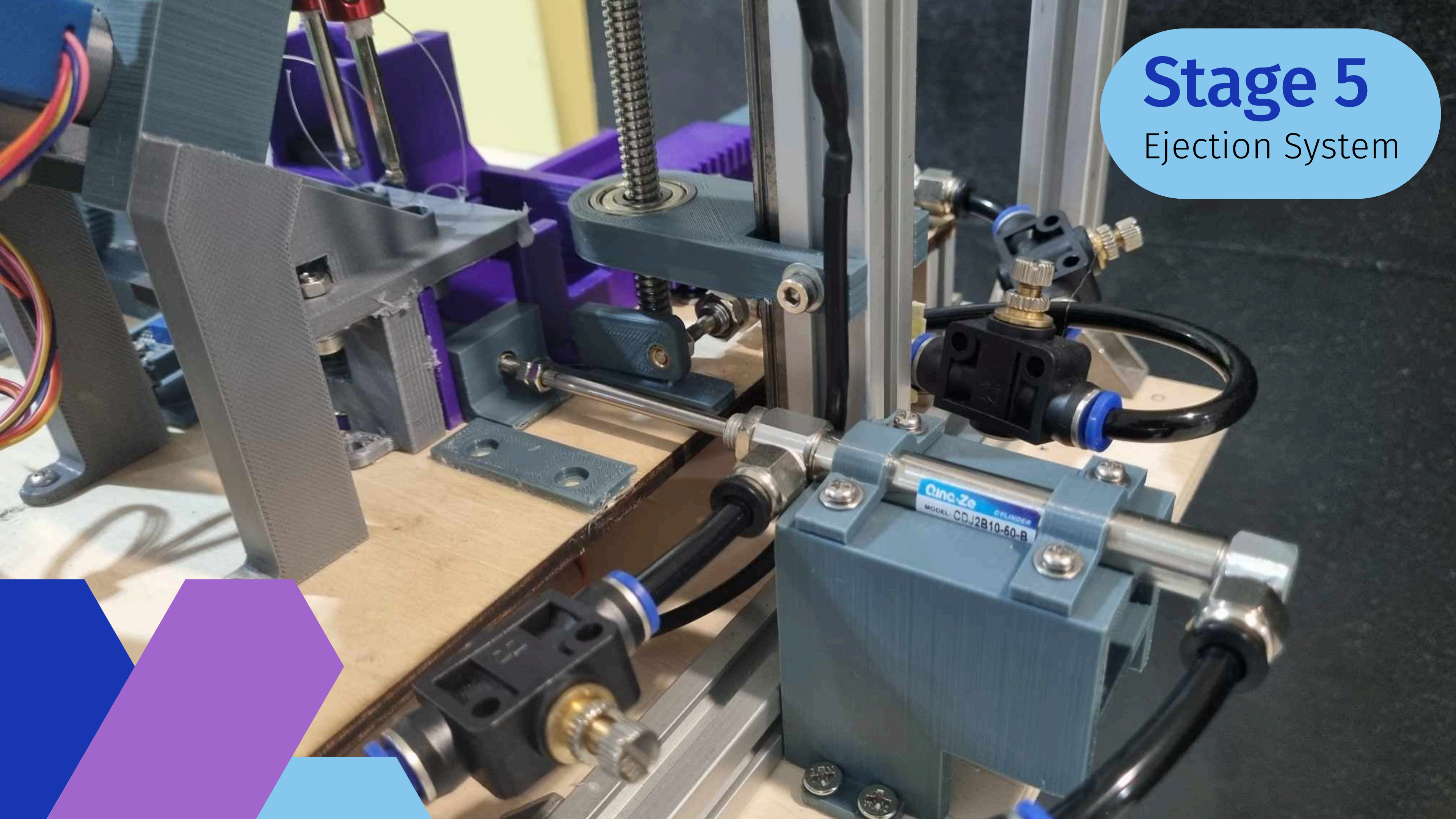
Stage 4

Screw
Tightening
system

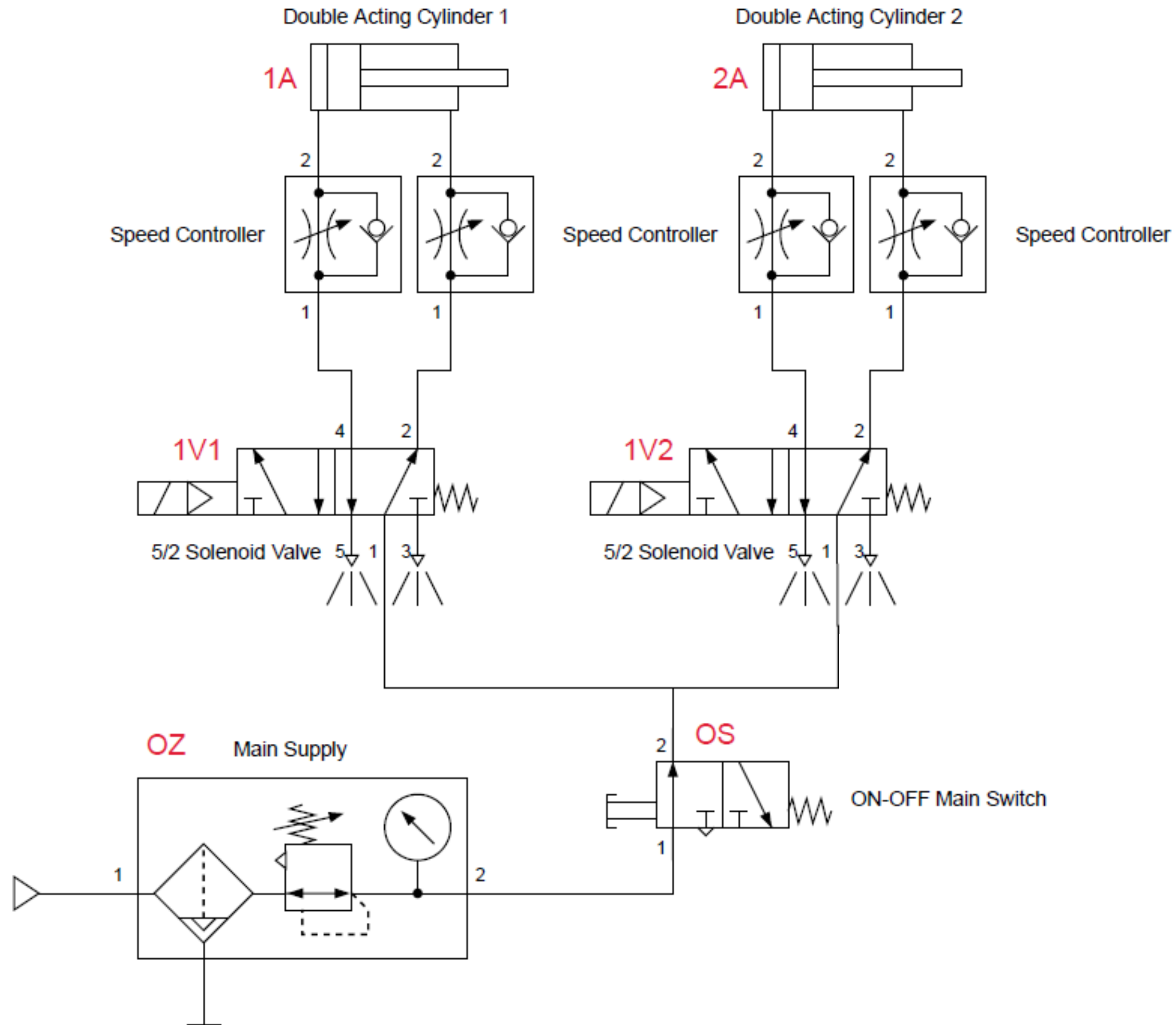


Stage 5

Ejection System



Pneumatic System



Electrical Circuit

